

Dr. Jing Yan HAW

PERSONAL PARTICULARS

Nationality: Singapore Permanent Resident (Malaysian)
Profiles: [Google Scholar](#), [LinkedIn](#)
Email: jy.haw@nus.edu.sg
Phone number: +6596580496



HIGHLIGHTS

- Main technical lead for Singapore's Quantum-Safe Network (>12 nodes with >15 use cases)
- Editor of the first Singapore's Reference Specification on quantum key distribution networks.
- Contributor to international standardisation (ITU-T, ETSI) of quantum technologies.
- Expert Researcher in quantum-safe technologies and quantum photonics engineering.
- Publications: Journals (19), Technical Reports (10), Books (3), Conferences (35), In prep. (4).

RESEARCH INTERESTS

- Quantum Communication Technologies
Quantum key distribution (QKD) networks, quantum random number generators (QRNG), hybrid quantum-safe architectures, post-quantum cryptography (PQC) deployment
- Quantum Photonics Engineering
Free-space, fibre-based & integrated quantum photonic platform for discrete & continuous
Variable quantum state generation & detection

ACADEMIC QUALIFICATIONS

Doctor of Philosophy in Physics 2013 - 2018
The Australian National University, Australia
• Thesis: *Continuous Variable Optimisation of Quantum Randomness and Probabilistic Linear Amplification* (Main supervisor: Prof. Ping Koy Lam)

Bachelor of Science (First Class Honours) in Physics 2007 - 2011
National University of Singapore (NUS), Singapore

PROFESSIONAL EXPERIENCE

Senior Research Fellow 2024 - Present
Quantum Communications Technologist 2022 - 2024
Centre for Quantum Technologies, National University of Singapore, Singapore

- Leading the deployment of a vendor-neutral, multi-protocol quantum-safe testbed with over >12 nodes of >1500km deployed fibre.
- Spearheaded QKD deployment at [National Quantum-Safe Network](#), including the [first QKD link between data centers in Singapore](#) and a [QKD-secured VPN for edge-cloud computing applications between NUS & Horizon Quantum Computing](#)
- Developed the [first Singapore local standard](#) on the deployment of QKD Networks & driving the first international standardisation for [QKD protocol framework](#) at ITU-T.
- Organising a high-performance, customized, centralized key manager system at the central node to ensure interoperability, connectivity, and scalability of the QKD network.
- Developing and overseeing the application layer within the NQSN QKD network, facilitating the integration and deployment of more than 15 diverse applications & proof of concepts

Research Fellow 2020 - 2022
Electrical & Computer Engineering, National University of Singapore, Singapore

- Conducted research on secure quantum randomness generation and broadband balanced quantum homodyne detection for GHz quantum communication
- Investigated hybrid discrete-continuous variable quantum key distribution

Independent Consultant
Berlin Germany

2018 - 2020

- Led the advisory and technical support for quantum communication experiments & quantum network at Fraunhofer Heinrich-Hertz Institute (HHI)
- Offered scientific consultation service: research methodology & technical analysis

Research Associate

2017

CQC2T, The Australian National University, Australia

- Engaged in machine learning cryptanalysis of a quantum random number generator
- Theory lead for the first integrated photonic chips for continuous variable QKD project

Research Assistant

2011 - 2012

Centre for Quantum Technologies, National University of Singapore, Singapore

- Investigated ultrastrong coupling in superconducting circuit system

Research Internship

2011

UTRIP, University of Tokyo, Japan

- Studied open quantum system dynamics with master equations

TEACHING EXPERIENCE

INVITED SEMINARS & LECTURES

1. Guest Lecturer (x4), Quantum Information Graduate Course PH7021 (School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore, 2022-2025)
2. Invited Speaker, Singapore University of Technology and Design: Science, Mathematics and Technology Research Seminar (Singapore, Zoom, 2021)
3. Guest Speaker, Humboldt University of Berlin (Germany, 2018)
4. Guest Speaker, TU Delft (The Netherlands, 2016)
5. Contributed Speaker, CQC2T Kioloa Optics Workshop (Australia, 2014)

RESEARCH SUPERVISION/CO-SUPERVISION

1. RESEARCHERS

- (i) Research Engineers/Assistants:
Ruyi Xu (2025, Quantum Technical Officer), Jeff Lee (2025, Quantum-Safe Application Engineer), Matthew Wee (2023-Present, Quantum Network Engineer), Cassey Liang (2020-24, Quantum-safe Software Engineer & Research Engineer), Romain Frappier (2024-25, Quantum-safe Software Engineer), KaiWei Qiu (2023-25, Quantum Network Intern & Project Officer), Raymond Ho (2020–22, Research Engineer), Fabian Zwiehoff (2019, Research Assistant)
- (ii) Interns:
Juyoung Shin (2025) Colin Hong (2023), SiHyung Kim (2023), Isaac Ng (2023)

2. STUDENTS

- (i) PhD Students:
Kaiwei Qiu (2025-Now), Ng Hong Jie (2020–2022), Tan Xiaojie (2020–2021), Thibault Michel (2017–2018), Hao Jeng (2017–2018)
- (ii) Undergraduate Students:
SiHyung Kim (2024-25), KaiWei Qiu (2022–23, Final Year Project), Kenny Kok, Beverley Goh, Ding Chao Wang, Joshua Yong (2020–21, Joint Capstone Project), Tang JinCheng (2020), Shiromal Kumara (2017)

LAB DEMONSTRATOR

1. The Australian National University (Jan 2014 - Jul 2016)
Supervision of undergraduate physics lab and evaluation of student reports
2. National University of Singapore (Aug 2010 - May 2011)
Conducted laboratory session of freshman physics courses

PROFESSIONAL & PUBLIC SERVICES

PROFESSIONAL AFFILIATIONS

1. Founding Committee Member of Singapore Computer Society Quantum Technology SIG
2. Editor for IMDA TSAC FA7 Secure Deployment of QKD Networks Task Force
3. Editor for IMDA TSAC FA7 Quantum Communication Networks Task Force (QCNTF)
4. ITU-T (SG 17 & JCA-QKDN) Contributor & Delegate
5. ETSI Industry Specification Group QKD Contributor & Member
6. Founding member of ETSI Technical Committee Quantum Technologies

WORKSHOP/EVENT ORGANISER

2026: National Quantum Office Quantum Industry Day
2025: Monetary Authority of Singapore Quantum-Safe Workshop; NQSN Workgroup Meeting; 10th ETSI/IQC Quantum Safe Cryptography Conference
2024: 4th meeting of Joint Coordination Activity (JCA) on Quantum Key Distribution Network
2022: ITU-T workshop; SG Innovate Events; Workshop Series: [Temasek Foundation - National University Of Singapore Youths For Singapore: Building A Shared Future - What is Quantum Computing?](#)

PEER REVIEW FOR JOURNALS

Nature Communications, Communications Physics, Physical Review Letter, Physical Review Applied, Quantum, Quantum Science and Technology, Advanced Quantum Technologies, Optica, Optics Express, SoftwareX, Entropy, Quantum Reports, Acta Astronautica

CONTRIBUTED & INVITED TALKS

1. Invited Speaker, French Singaporean Quantum Symposium (France, 2025)
2. Contributed Speaker, [PKI Consortium PQC Conference](#) (Malaysia, 2025)
3. Invited Speaker, SCS Quantum Technologies SIG, NQSN Lab Tour (Singapore, 2025)
4. Contributed Speaker, QCrypt 2025 (China, 2025)
5. Invited Panelist, APEC Workshop on Quantum Standardisation (Korea, 2025)
6. Invited Speaker, Fortinet AI & Quantum-Safe Security: Future-Proofing the Network (SG, 2025)
7. Invited Speaker, Singapore Computer Society (SCS) Quantum Technologies SIG (Singapore, 2025)
8. **Keynote Speaker**, Sunway University Quantum Day 2025 (Malaysia, 2025)
9. Invited Panel Speaker, GITEX Asia (Singapore, 2025)
10. Invited Speaker, TUM Campus Heilbronn – Global Technology Forum (Heilbronn, 2024)
11. Contributed Speaker, 10th ETSI/IQC Quantum Safe Cryptography Conference (Singapore, 2024)
12. Contributed Speaker, European Conference on Optical Communications (Glasgow, 2023)
13. Invited Speaker, Supercomputing Asia 2023 (Singapore, 2023)
14. Invited Speaker, IMDA Brown Bag - Quantum Communications (Singapore, 2023)
15. Invited Speaker, First Singapore Quantum Industry Day (Singapore, 2023)
16. Invited Speaker, [SGInnovate - Quantum-Safe Cybersecurity](#) (Singapore, 2022)
17. Panel Moderator [SGInnovate - What Cybersecurity Needs To Get Quantum-Safe](#) (Singapore, 2022)
18. **Keynote Speaker**, Integrated Health Information Systems (IHiS) Cybersecurity Week 2022 (Singapore, 2022)
19. Invited Speaker, Student Conference on Optics and Photonics (SCOP) 2022 (Physical Research Laboratory, Ahmedabad, India, 2022)
20. Contributed Speaker, Institute of Physics Singapore 2021 Meeting (Singapore, 2021)

21. Contributed Speaker, QCrypt, 10th international conference on quantum cryptography (The Netherlands, 2020)
22. Contributed Speaker, 46th European Conference on Optical Communication (Belgium, 2020)
23. Contributed Speaker, Asia Pacific Physics Conference and 22nd Australian Institute of Physics Congress (Australia, 2016)
24. Contributed Speaker, 21st International Congress of the Australian Institute of Physics (Australia, 2014)

RESEARCH GRANTS AWARDED/INVOLVED

1. CQT Young Researcher Career Development Grant
 - Total awarded: SGD 26,000 (2025-2026), Host: CQT
 - Role: Principal Investigator for "RaspiQRNG – A DIY, open Quantum Random Number Generator based on Raspberry Pi" Project
2. Provision of Quantum Consultancy and Support Services to IMDA
 - Total awarded: SGD 999,600 (2025-2027), Host: CQT
 - Roles: Led the organisation the scope of work, deliverables and budget (manpower, expenses) for the consultancy contract to support IMDA's NQSN plus initiative
3. Quantum Engineering Programme 3.0: National Quantum-Safe Network Testbed 2025-2029
 - Total awarded: SGD 3,120,000 (2025- 2029). Lead: Prof Alexander Ling
 - Roles: (1) Participation in the drafting the grant proposal, industry engagement & budgeting; (2) Direct reporting to the National Quantum Office on the project status
4. Quantum Engineering Programme 2.0: National Quantum-Safe Network
 - Total awarded: SGD 8,499,600 (2021-2025). Lead: Prof. Alexander Ling
 - Roles: (1) Participation in the initial planning of the grant proposal; (2) Coordination of the utilisation of the funds within the project, including manpower, equipment and other operating expenses.

ACHIEVEMENTS & MEDIA

1. [Compact quantum randomness generator emerges from undergrad project](#), CQT Highlight (2026)
2. [Bright photonic chip could power quantum networks](#), CQT Highlight (2025)
3. [CQT updates roles in national quantum programmes](#), CQT Highlight (2025)
4. [Exploring the Future of Quantum-Safe Networks at TM R&D's Big Idea Lecture Series](#), TM R&D News & Events (2025).
5. [Singapore announces National Quantum Strategy](#), CQT Highlights (2024)
6. [CQT hosts the 10th ETSI/IQC Quantum Safe Cryptography Conference](#), CQT Highlights (2024)
7. [Driving towards standards for quantum key distribution](#), CQT Highlights (2023)
8. [Singapore data centres exchange quantum-safe encrypted files](#), CQT Highlights (2023)
9. [Events build awareness for National Quantum-Safe Network](#), CQT Highlights (2023)
10. [Implementing a quantum-secured network in a metropolitan area](#), AWS QT Blog (2023)
11. [First Singapore Quantum Industry Day](#), National Quantum Office Newsroom (2023)
12. [Quantum technologies feature in NUS programme for schools](#), NUS News & CQT Highlights (2022)
13. [Meet a CQTian: Haw Jing Yan](#), CQT Highlights (2022)
14. [Narrowing the gap between theory and practice for quantum secure communications](#), NUS News & CQT Highlights (2021)

15. [Precise quantum cloning: Possible pathway to secure communication](#), Phys.Org (2016)
16. [The Straits Times reports on textbook co-authored with students](#), CQT Highlights (2010)

AWARDS

1. The Australian National University PhD Scholarship (2013 - 2017)
2. University of Tokyo Research Internship Programme (2011)
3. A*STAR Pre-Graduate Scholarship (2010 - 2011)
4. NUS Faculty of Science Dean's List (2009, 2010)

TECHNICAL SKILLS

COMPUTER PROGRAMMING

Python, MATLAB, Mathematica, LabView, LaTeX, PHP and MYSQL

LANGUAGES

Mandarin (Native), English (Fluent), Malay (Fluent), German (Elementary), Japanese (Basic)

REFEREES

1. Prof. Alexander Ling Euk Jin
Professor, Centre for Quantum Technologies, National University of Singapore
phyalej@nus.edu.sg
2. Prof. Ping Koy Lam
A*STAR Chief Quantum Scientist & Professor of Physics, Australian National University
Email: pingkoy@a-star.edu.sg
3. Assoc. Prof. Charles Lim
Associate Professor, National University of Singapore
Email: charleslim.research@gmail.com
4. Prof. Antia Lamas-Linares
Principal Quantum Scientist, AWS Core Networking & Visiting Research Professor, Vigo Quantum Communications Center
Email: alamasli@amazon.com
5. Mr. Koh Wee Sain
Cluster Director, Engineering, Infocomm Media Development Authority
Email: koh_wee_sain@imda.gov.sg